Delivered tables — preview

Not a deliverable. Layout here is arbitrary; the paper's floats are D's call. Generated by `python -m analysis.preview`.

table1_dataset

Statistic	Development	Test
Paragraphs	2000	2000
Source reports (PDFs)	49	49
shared across splits	49	49
Companies	49	49
Legal states observed	15 / 17	n/a
Rarest classes		
within_2_years	34	n/a
Misleading	2	n/a

Dataset and split statistics, regenerated from the versioned data by analysis/audit.py. Misleading occurs twice in the whole development set, in two different reports, and is absent from the Calibration partition of 18 of the 30 rotations. The four fields admit 120 combinations, of which the hierarchy leaves 17 legal, and no development row violates it. All 49 development reports also appear in the test split, and one paragraph text is duplicated across the two. No company contributes more than one report, so a document-disjoint split is also company-disjoint. The competition test split ships no labels, so its label-derived cells are marked n/a.

table2_main

ID	Calibration	Decoding	Weighted F1	PS	VT	ES	EQ	Tuple Acc.	Invalid %
M0	None	Independent	0.572±0.003	0.796	0.464	0.650	0.424	0.359	12.6
M1	None	Projection	0.571±0.003	0.796	0.467	0.651	0.418	0.394	0.0
M2	Global	Projection	0.574±0.004	0.796	0.493	0.650	0.418	0.428	0.0
M3	Conditional	Projection	0.574±0.005	0.796	0.501	0.651	0.412	0.431	0.0
M4	None	17-state	0.571±0.005	0.799	0.468	0.648	0.420	0.388	0.0
M5	Global	17-state	0.576±0.001	0.796	0.493	0.649	0.423	0.430	0.0
M6	Conditional	17-state	0.567±0.009	0.784	0.494	0.636	0.414	0.426	0.0

Controlled comparison of decision rules on identical base probabilities and identical test rows. Each cell is the mean over three seeds of a single score computed on all 2,000 concatenated test rows; ± is the sample standard deviation across seeds, which reflects the variability of the whole pipeline – fold assignment and training together – rather than model stability. Paired contrasts between these rows, under each metric and with their Holm-corrected p-values, are reported in Table 4. Under that variant M0 scores 0.567 against 0.572 on the official metric, while M1–M6 score identically under both: only the unconstrained arm predicts fields its own ancestors do not support. The metrics disagree about the winner: M5 on the official metric and M6 on hF, which is the lowest-scoring rule on the official metric. A ranking is only meaningful stated together with the metric that produced it.

table3_regimes

Method	Same-document	Document-disjoint	Δ	95% CI
M0	0.585	0.572	0.012	[0.004, 0.022]
M1	0.581	0.571	0.010	[0.001, 0.019]
M2	0.587	0.574	0.012	[0.001, 0.024]
M3	0.588	0.574	0.014	[0.003, 0.025]
M4	0.586	0.571	0.014	[0.005, 0.024]
M5	0.590	0.576	0.015	[0.004, 0.027]
M6	0.587	0.567	0.020	[0.011, 0.031]

Same-document versus document-disjoint evaluation. The left column measures seen-report, unseen-paragraph generalisation and matches the competition distribution, since test and development draw on the same 49 reports; the right column measures generalisation to entirely unseen reports. Δ is the gap between two estimation targets and is not a bias estimate. All seven decision rules are reported rather than a best-scoring subset: every Δ is positive and every interval excludes zero, so the gap is a property of the split and not of any one rule.

table4_contrasts

Metric	Contrast	Δ	95% CI	p_{Holm}
Weighted macro-F1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	-0.001	[-0.006, 0.004]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.007	[-0.017, 0.003]	0.643
	M6-M5	-0.009	[-0.017, -0.001]	0.135
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.002	[-0.003, 0.007]	0.973
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.014, 0.004]	0.973
	M6-M5	-0.004	[-0.012, 0.004]	0.973

The five pre-specified contrasts of the analysis plan, under the two metrics the plan named before any result existed: the competition's own weighted macro-F1 and tuple exact-match. Paired PDF-cluster bootstrap over 10,000 resamples on the document-disjoint protocol; the resampling unit is the source report, not the paragraph. Each metric is Holm-corrected as its own family of five rather than pooled: the contrasts were specified once, not once per metric. **Bold p_Holm marks a contrast that survives the correction; the bracketed intervals are uncorrected 95% percentile intervals and decide nothing on their own.** Two further structure-aware metrics were examined after the primary analysis and are reported as exploratory in the text, not as Holm families here. The largest effect here, M6-M5 at $|\Delta| = 0.009$, has an interval excluding zero and still does not survive the correction (p_Holm = 0.135); for scale, the full range across the seven decision rules is 0.0088. Of the five contrasts, none on Weighted macro-F1 (official) and two on Tuple accuracy survive the correction at α=0.05. The official metric and tuple accuracy were named in the analysis plan in advance; the path-constrained variant and the hierarchical F1 were adopted after the primary analysis and are not pre-specified.

table8_headroom

Field	Weight	macro-F1	Shortfall	Share	Value of +0.1 F1
evidence_quality	0.35	0.418	0.204	47.5%	0.0087
evidence_status	0.30	0.651	0.105	24.4%	0.0100
verification_timeline	0.15	0.467	0.080	18.6%	0.0030
promise_status	0.20	0.796	0.041	9.5%	0.0100

Where the official score falls short under M1, and what closing each part is worth. Shortfall is the field's weight times its distance from a perfect macro-F1; share is its portion of the total. The last column is what a 0.1 gain in that field's macro-F1 adds to the weighted total, which orders the fields differently from the shortfall: macro-F1 averages over classes, so a field carrying more of them returns less per unit gained. **Not all of the shortfall is reachable.** Misleading occurs twice in the development set and is never predicted by any arm, so a quarter of evidence_quality's macro-F1 – 0.0875 of the weighted total – is out of reach by construction.

table6_legality_cost

Backbone	λ	M0 invalid	Enforce legality (M1−M0)		Joint decoding (M4−M1)	
			Official	Whole-row	Official	Whole-row
RoBERTa-large	0	12.55%	-0.001	**+0.035**	+0.000	**-0.006**
DeBERTa-v2-320M	0	19.75%	**-0.008**	**+0.050**	**+0.013**	-0.003
ELECTRA-large	0	10.20%	**-0.005**	**+0.024**	**+0.007**	-0.000
RoBERTa-base	0	11.77%	**-0.007**	**+0.030**	**+0.008**	-0.003
RoBERTa-large	0.3	5.18%	-0.001	**+0.014**	-0.001	-0.001
DeBERTa-v2-320M	0.3	6.53%	+0.001	**+0.015**	-0.002	**-0.003**
ELECTRA-large	0.3	3.68%	-0.000	**+0.009**	-0.000	-0.002

Two decision rules over one set of probabilities, on the pdf_group protocol. M0 takes each field's argmax independently; M1 projects onto the 17 legal states top-down; M4 scores all 17 and takes the best. **M1 and M4 both emit an invalid tuple on 0% of rows in every arm** – their output space is the legal set – so the invalid column reports M0 only, where it ranges from 3.68% to 19.75%. Means over 3 seeds; paired PDF-cluster bootstrap, 10,000 resamples, seed 20260814, one resample shared within each contrast. Enforcing legality raises whole-row accuracy in all 7 arms and lowers the official score in all 4 arms trained without the structural objective; joint decoding reverses both signs. The 3 structurally trained arms show neither effect. **Exploratory.** These contrasts were named after the primary analysis and form no Holm family; bold marks an interval excluding zero, not a corrected verdict. The claim is the sign pattern across arms, not any single cell. See docs/inference_families.md.

figure1_hierarchy